Cost vs. quality: when can cheap heterogeneous models replace one expensive model?

A practical question with a clean answer from the competence-band theory: given the accuracies of cheaper models on a class of questions, can a panel of them replace a single costly frontier model? **It depends on whether the answer is verifiable.**

The data (100 hard questions: TruthfulQA + MMLU-Pro)

Individual accuracy:

Model	Accuracy	Relative cost ($/1M out)
claude-opus (expensive)	**92%**	~75
claude-sonnet	78%	~15
qwen235b	75%	~3
gpt-5 (min-reasoning)	69%	~10
gpt-4o	66%	~10
deepseek-v3	65%	~1
llama70b	58%	~1

Cheap panel vs. expensive single:

Approach	Majority vote	Coverage (verify ceiling)
Opus alone	**92%**	—
cheap panel (gpt4o+deepseek+qwen235b)	69%	84%
5 cheap models	68%	87%
all 6 non-opus	**78%**	**91%**

The reasoning

Why majority vote of cheap models does NOT reach Opus. Majority voting can only amplify what the models already agree on. Because LLM errors are correlated (ρ ≈ 0.5), the cheap models tend to be wrong *together*, so the vote lands at roughly the **best single cheap model** (78% = claude-sonnet) and no higher. Adding more cheap models does not help — they bring correlated mistakes and dilute the strongest member. **Blind voting cannot turn cheap models into an expensive one.**

Why coverage (verify) almost DOES reach Opus. The cheap models *fail on different questions* enough that *at least one* of them is right **91%** of the time — within a point of Opus's 92%. That coverage is a real, latent resource. The catch: to cash it in you must be able to **identify the correct answer when it is present** — i.e. *verify*.

The decision rule. - **Verifiable task** (code with tests, math with a checker, or a reliable judge): a panel of cheap models + a verifier reaches ≈ the coverage (~91%), **matching Opus at a fraction of the cost** (a handful of cheap calls < one Opus call). **Use the cheap panel.** - **Non-verifiable task** (must trust the answer, majority vote): the panel lands at the best single cheap model (78%), well below Opus. **Use the single best model you can afford** — a panel buys nothing.

This is the practical face of the competence-band theory: diversity creates a *coverage ceiling* (set by error correlation), and **the aggregator — i.e. whether you can verify — decides how much of that ceiling you capture.**

Empirical confirmation (verifiable STEM subset) ✅

MMLU-Pro math/physics/chemistry/engineering (40 questions; objectively checkable):

Approach	Accuracy	Rel. cost / 1000 Q
claude-opus alone (expensive)	**81%**	~$11
gpt-4o alone	48%	
deepseek-v3 alone	56%	
qwen235b alone	68%	
cheap panel [gpt4o+deepseek+qwen235b] + judge (debate, 2 rounds)	**81%**	**~$5**

Three cheap models — each individually far below Opus — **matched Opus (81% = 81%) at ~2× lower cost**. On STEM the cheap models are well-decorrelated (ρ=0.30, N_eff=1.89 of 3), giving 80% static coverage; debate + judge reached 81%, slightly exceeding static coverage because two rounds of deliberation also *improve* answers, not just select among them. **This is the verifiable case made concrete: cheap diverse panel + checker = an expensive single model, cheaper.**

A decision tool (the theory, made usable)

`code/debate/decision_tool.py` turns sample accuracies into a recommendation:

```
coverage ceiling   C    = 1 − (1 − ā)^N_eff      with N_eff = N/(1+(N−1)ρ)
projected panel    R    = a_best + v·(C − a_best)   (v: verifier quality, 0=blind, 1=oracle)
recommend cheap panel iff R ≥ a_expensive − tol  AND  cost_panel < cost_expensive
```

Worked from our 6 cheap models (accuracies 58–78%, ρ=0.48) vs Opus 92%: | scenario | verifier v | projected panel | recommend | |—|—|—|—| | non-verifiable (blind majority) | 0.0 | 78% (= best single cheap) | **expensive single** | | verifiable (strong checker) | 0.9 | 86% (analytical) / ~91% (empirical) | depends on Δ to Opus |

Caveat on the formula. The exchangeable estimate C = 1−(1−ā)^N_eff is a *conservative lower bound* (it assumes worst-case homogeneity within the effective voters); the *measured* coverage of the same 6 models was 91% vs the formula's 87%. Use the formula for planning and the empirical coverage when available.

The headline for practitioners: estimate your cheap models' accuracy and correlation on a sample of the target questions; if the task is **verifiable**, a cheap diverse panel + checker typically matches a costly single model at a fraction of the price; if it is **not verifiable**, pay for the best single model — a cheap panel will only reach the best cheap member.